



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

10-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A	A	A	C	B	C		
SECTION-I (ii)	Q.	7	8	9	10	11	12	13	14
	A.	A,B	A,C	A,C	B,C	B,C	C,D	A,B	A,C
SECTION-I (iii)	Q.	15	16	17	18				
	A.	A	B	B	C				

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	D	C	A	D	D	D		
SECTION-I (ii)	Q.	7	8	9	10	11	12	13	14
	A.	A,B,C	A,B,C,D	A,B,C	A,B,C	B,C,D	A,C,D	A,B,C	C
SECTION-I (iii)	Q.	15	16	17	18				
	A.	A	D	D	A				

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	D	C	C	D	C	B		
SECTION-I (ii)	Q.	7	8	9	10	11	12	13	14
	A.	A,B,C,D	A,B,C,D	B,C,D	A,B,C	A,C	B,C,D	C,D	B,D
SECTION-I (iii)	Q.	15	16	17	18				
	A.	A	B	A	A				

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A)**

$$F = mV^2 / R = 2mg \sin \theta$$

$$N(mV^2 / R) + mg \sin \theta = 3mg \sin \theta$$

$$\text{So, } F/N = 2/3 = \text{constant.}$$

2. **Ans (A)**

Writing torque equation for an arbitrary ' θ ' position shown in the diagram (Taking ACW sense as positive) $-mg\ell \sin \theta - mg\ell = I\alpha$.

Taking $\sin \theta \approx \theta$ for small angle given $-mg\ell$

$$(\ell + \theta) = m\ell^2 \alpha \Rightarrow \text{time period} = 2\pi \sqrt{\frac{\ell}{g}}$$

So equilibrium position is $\theta = -1$ radian had the force continued to be applied $\Rightarrow$ amplitude. So the present problem is equilateral to calculating

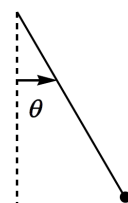
time taken to move from $\theta' = \frac{2}{\sqrt{3}}$ rad to $\theta' = 1$

(where θ' is angle measure from equilibrium position).

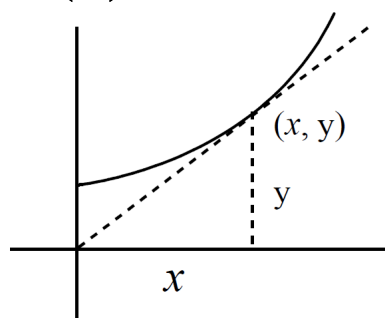
$\Rightarrow$ it is same as time taken to move from $a = A$

to $x = \frac{\sqrt{3}A}{2}$, which is equal to $\frac{T}{12}$

$$\Rightarrow \text{time taken} = \frac{1}{2} 2\pi \sqrt{\frac{\ell}{g}} = \frac{\pi}{6} \sqrt{\frac{\ell}{g}}$$



3. Ans (A)



Maximum distance = max (VT)

This happens when $\min \left(\frac{1}{VT} \right)$ occurs

$$= \min \left(\frac{1/T}{V} \right)$$

= min slope of a line connecting origin to a point on the curve.

= slope of a tangent from origin to the curve

$$\text{Given } \frac{y}{x} = \frac{0.04}{x} + x \geq 2\sqrt{0.04}$$

$$\Rightarrow \frac{y}{x} \geq 0.4$$

$$\Rightarrow \text{maximum distance} = \frac{1}{0.4} = 2.5\text{m}$$

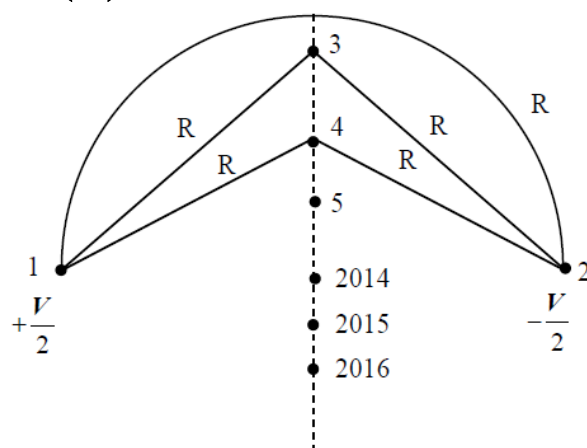
4. Ans (C)

We cannot use $\frac{\sum m_i v_i}{\sum m_i}$ for v_{cm} for a variable

mass system. As the balls are evacuated the

C.M of remaining balls keeps going down.

5. Ans (B)



Assume the points are numbered from 1 to 2016 and

battery is connected to points 1 and 2. Re-arrange

these points in a plane such that point 3 to 2016 lie

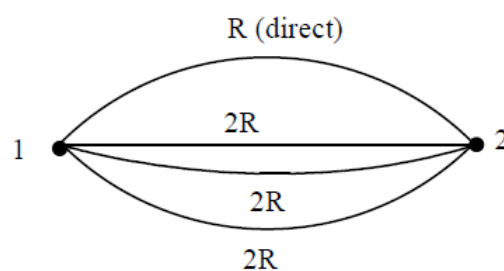
on perpendicular bisector plane of 1 and 2. Since

this plane is plane of symmetry all

resistors connecting 3 to 2016 (internally can be

removed. Now circuit becomes simple

series parallel one as shown.



$R_{eff} = R$ (Parallel to) (2014 parallel 2R's)

$$\Rightarrow \frac{1}{R_{eff}} = \frac{1}{R} + \frac{2014}{2R} = \frac{1008}{R}$$

$$\Rightarrow R_{eff} = \frac{1}{1008} \Omega$$

Alternately : For n points, you can use principle of

mathematical induction for $n = 2$, $n = 3$ and prove it

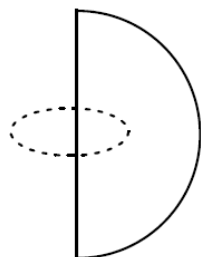
for n as $\frac{2R}{n}$.

6. **Ans (C)**

Complete the loop as shown. Due to total loop

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

Due to straight part $\int \vec{B} \cdot d\vec{l} = -\frac{\sqrt{3}}{2} \mu_0 i$



PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A,B)**

C and D should have same polarity of connections

8. **Ans (A,C)**

For a small displacement x , $Mg - T = Ma$

$$T - m(R - x)(\omega')^2 = ma$$

$$\Rightarrow Mg - m(R - x)(\omega')^2 = (m + M)a$$

Also $m(R - x)^2 (\omega') = m(R)^2 \omega$ from COAM

$$\Rightarrow Mg - \frac{mR^4 \omega^2}{(R - x)^3} = (m + M)a$$

$$Mg - mR\omega^2 \left(1 + \frac{3x}{R}\right) = (m + M)a \quad [\text{for } x \ll R]$$

$$a = -\frac{3m\omega^2}{m + M}x$$

$$\text{Net force on } M = ma = \frac{3mM\omega^2 x_0}{2(M + m)}$$

$$\text{Net force on } m = T = Mg + \frac{3x_0 m M \omega^2}{2(M + m)}$$

9. **Ans (A,C)**

In the flowing river frame position vector of coordinate of one flower with respect to other should give us v_{br} , which could be $\pm (4\hat{i} + 3\hat{j})$ m/s depending which flower was dropped first.

$$\text{Thus } \vec{v}_b = \vec{v}_{br} + \vec{v}_r = \pm (4\hat{i} + 3\hat{j}) + (-3\hat{i})$$

10. **Ans (B,C)**

$$M.E = (U + K) \text{ at } x = 1\text{m}$$

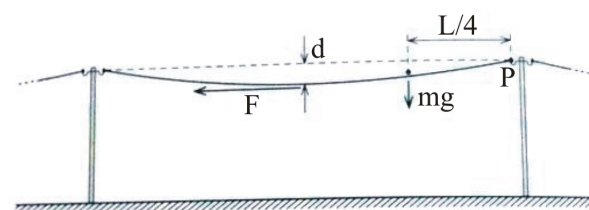
$$= 3 + \frac{1}{2} \times 2 \times 1^2 = 4\text{J}$$

But $U = 4\text{ J}$ happens at $x = 6$. This is the position where particle reverses its direction of motion.

Force at $x = 6$ is given by $-\frac{\partial U}{\partial x}$, which is negative.

11. **Ans (B,C)**

For the equilibrium of any arbitrary piece of the cable, both the net force and the net torque acting on that piece need to be zero. Consider, therefore, the equilibrium of the segment that has a suspension point P and the point of 'maximum sag' as its two ends.



Because of the relatively small sag, the length of the selected piece can be taken as $L/2$ and its mass as $m = \lambda L/2$. Among the various forces acting on this line segment, only the horizontal tension F (exerted by the neglected half of the cable) and the gravitational force mg produce torques about the point P; the (almost) horizontal reaction force at P produces no torque about that point. The moment arm of force F is d , while that of the weight can be taken as $L/4$ (also because $d \ll L$), and consequently, as the net torque must be zero, $Fd - mg \frac{L}{4} = 0$.

Inserting the expression for the mass into this equation gives the approximate magnitude of the tension in the cable as $F = \frac{\lambda L^2 g}{8d}$.

At the same level as the other approximations already made, this can be taken as the tension throughout the cable.

12. Ans (C,D)

We see that 49 MSD = 50 VSD

$$\Rightarrow L.C = \frac{1}{50} \text{MSD} = 0.01 \text{mm}$$

If 24th division is coinciding exactly.

$$\text{Reading} = (51 \text{ mm}) + 24 (L.C) = 51.24 \text{ mm.}$$

But 24th division is just to the right of MSD. So

actual length is greater than 51 mm + 24 (L.C)

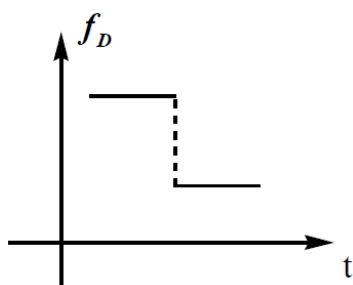
but less than 51 mm + 25 (L.C)

13. Ans (A,B)

Flow will be steady W.R.T a frame attached to the submarine.

14. Ans (A,C)

The closer the detector is, the graph becomes more abrupt. If D is on P then the graph is as follows



Also f_D at $t_0 = \left(\frac{v}{v - u \cos \theta} \right) f$ where $\cos \theta = \frac{u}{v}$.

This f_D is independent of distance of D from P.

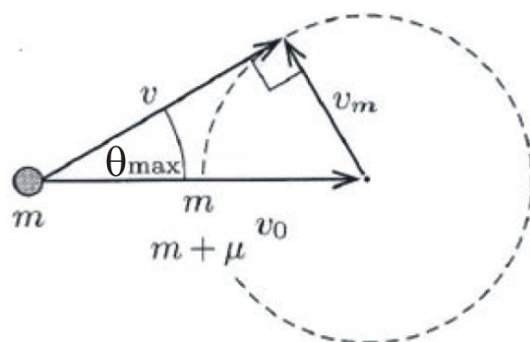
PART-1 : PHYSICS

SECTION-I (iii)

15. Ans (A)

If the mass ratio M/m is sufficiently large, then it can always be arranged, using fine tuning of the impact parameters (and some good luck!), that D will bounce from each of the discs of mass $\mu = M/N$ in turn. To a good approximation, its direction is monotonically changed by an angle of π/N at each bounce. However, if the common mass μ of the discs initially at rest is smaller than a certain critical value, then it is not possible, even with ideal impact parameter, to make D turn through an angle of π/N at each collision.

We first investigate the maximal angle through which a disc of mass m can be scattered by another with mass μ and initially at rest from the below vector diagram.



We can deduce that largest possible angle, $\theta_{\max}$, through which disc D can be scattered is given

$$\text{by } \sin \theta_{\max} = \frac{\mu}{m} = \frac{M/N}{m}. \quad (2)$$

A successful realization of the miraculous motion of the disc is possible if $\theta_{\max} \geq \pi/N$. As we are investigating the limiting case of $N \rightarrow \infty$, and $\theta_{\max}$ is therefore very small, the approximation $\sin \theta_{\max} \approx \theta_{\max}$ can be used, and the required condition on the masses involved becomes

$$\frac{\pi}{N} \leq \frac{M/N}{m} \text{ from which } \frac{M}{m} \geq \pi.$$

16. Ans (B)

Henceforth, only the case of critical mass ratio, i.e. when $M/m = \pi$, will be investigated. It can be seen from figure that, for the maximal angular deviation, the connection between the speeds of disc D before and after the collision can be expressed as follows:

$$v^2 = \left(\frac{m}{m + \mu} \right)^2 v_0^2 - v_m^2 = \frac{m^2 - \mu^2}{(m + \mu)^2} v_0^2.$$

Because $\mu \ll m$, speed v can be approximated as follows

$$\begin{aligned} v &= m \left(1 - \frac{\mu^2}{m^2} \right)^{1/2} m^{-1} \left(1 + \frac{\mu}{m} \right)^{-1} v_0 \\ &= \left(1 - \frac{\mu}{m} + \frac{\mu^2}{m^2} - \frac{1}{2} \frac{\mu^2}{m^2} + \dots \right) v_0 \\ &\approx \left(1 - \frac{\mu}{m} \right) v_0 \end{aligned}$$

So the speed of the disc decreases by a factor of $(1 - \mu/m)$ at each bounce, and the ration of the final to initial speed (after N bounces) is

$$\frac{v_{\text{final}}}{v_{\text{initial}}} = \left(1 - \frac{M}{Nm} \right)^N \approx e^{-M/m} = e^{-\pi} \approx 0.043.$$

17. Ans (B)

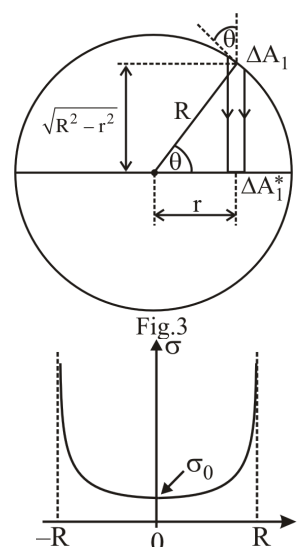
The electric field produced at P by the two charged pieces (projected onto the plane, and still glued to the disc) can be calculated from Coulomb's law:

$$|E_1^+ + E_2^+| = k_e \frac{\Delta Q_2}{d_1^2} - k_e \frac{\Delta Q_2}{d_1^2} = k_e \sigma_0 \left(\frac{\Delta A_1}{d_1^2} - \frac{\Delta A_2}{d_2^2} \right) = 0.$$

In the last step, the geometrical results $\frac{\Delta A_1}{\Delta A_2} = \frac{r_1^2}{r_2^2}$ and $\frac{r_1}{r_2} = \frac{d_1}{d_2}$ were used. The same zero-field results holds for all other charge pairs, and so also for the whole charge distribution; the electric field is everywhere zero on the plane of the disc.

17. Ans (B)

We have obtained the interesting result that the electric field produced by all the other charges does not produce any force on the charge fixed at P, for any P. Consequently, we can even abolish the notional fixing (gluing) of the charges, because – in the absence of any force - they will not move. The question of the electrical charge distribution on the disc has thus been solved; the charge is distributed on the disc in the same way as it would be if that on a uniformly charged spherical surface were projected perpendicularly onto it.



The surface charge density can be found with the help of a 'side view' sketch of a sphere of radius R , in which the particular surface pieces appear as line segments. The charges, originally situated on the area ΔA , are, after projection, at a distance r from the centre of the disc of radius R , and occupy an area $\Delta A^* = \Delta A \sin \theta = \frac{\sqrt{R^2 - r^2}}{R} \Delta A$ at their new position. So the charge density on the disc at a distance r from its centre is

$$\sigma(r) = \frac{\Delta Q}{\Delta A^*} = \sigma_0 \frac{\Delta A}{\Delta A^*} = \frac{Q}{4\pi R^2} \frac{R}{\sqrt{R^2 - r^2}}$$

18. Ans (C)

The electric field produced at P by the two charged pieces (projected onto the plane, and still glued to the disc) can be calculated from Coulomb's law:

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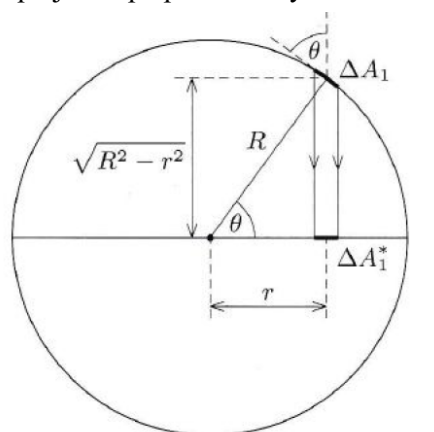
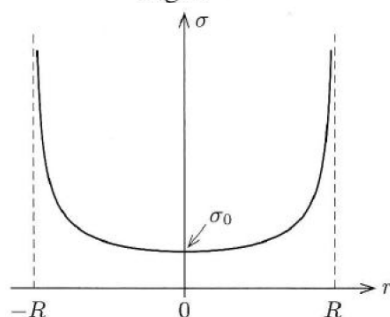


Fig. 3



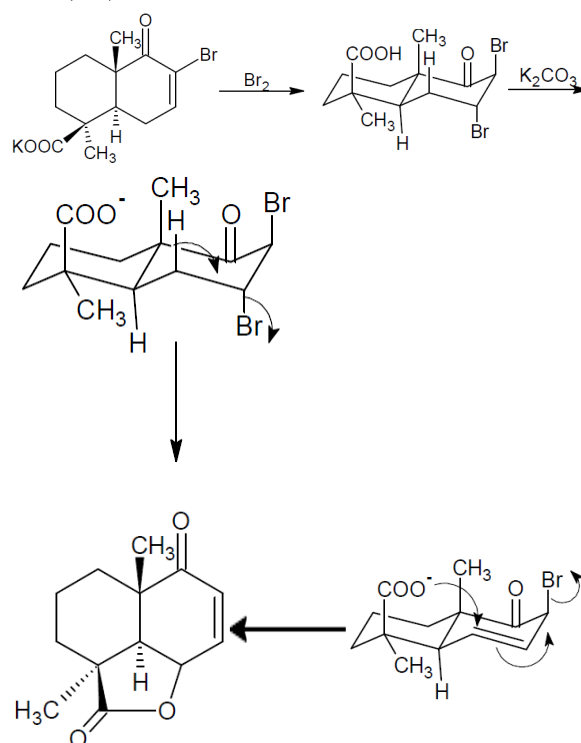
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$$\sigma(r) = \frac{\Delta Q}{\Delta A^*} = \sigma_0 \frac{\Delta A}{\Delta A^*} = \frac{Q}{4\pi R^2} \frac{R}{\sqrt{R^2 - r^2}}$$

PART-2 : CHEMISTRY

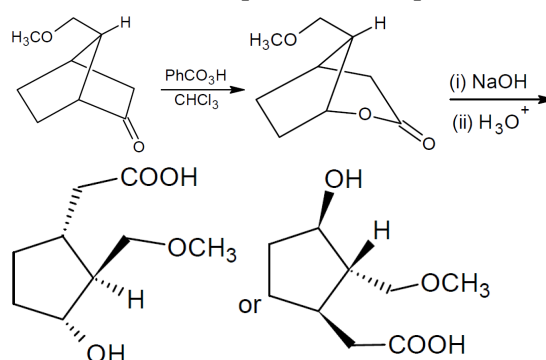
SECTION-I (i)

1. Ans (D)



2. Ans (C)

Anti addition takes place to the triple bond.



4. **Ans (D)**

With high concentration of nucleophile reaction is S_N2 and gives inverted product. At low concentration due to NGP it gives a product with retention of configuration.

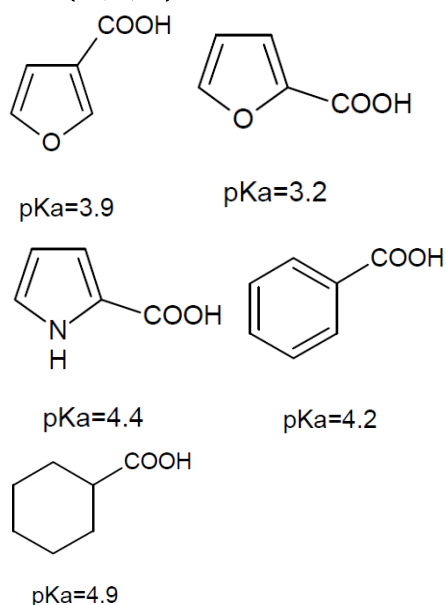
5. **Ans (D)**

Conceptual

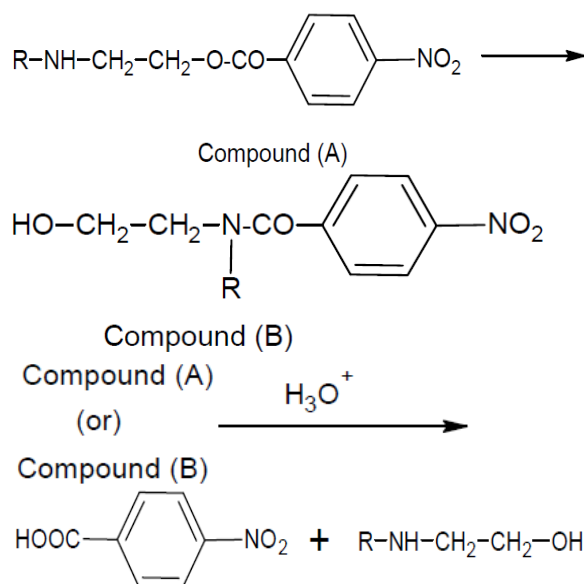
PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A,B,C)**



8. **Ans (A,B,C,D)**



9. **Ans (A,B,C)**

Conceptual

10. **Ans (A,B,C)**

Conceptual

11. **Ans (B,C,D)**

Conceptual

12. **Ans (A,C,D)**

Conceptual

13. **Ans (A,B,C)**

Conceptual

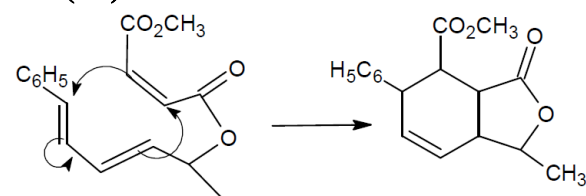
14. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-I (iii)

15. **Ans (A)**



16. **Ans (D)**

Conceptual

17. **Ans (D)**



BrSbF_8 ionises as BrF_2^+ & SbF_6^- and increases the concentration of $[\text{BrF}_2^+]$ hence it is an acid where as KBrF_4 ionises to give BrF_4^- hence it is a base.

18. **Ans (A)**

Conceptual

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$z^{n-1} = (z-1)(z^{n-1} + z^{n-2} + \dots + z + 1)$$

2. **Ans (C)**

The equation of the normal at $p(\alpha)$ on $x^2 + 4y^2 = 4$

is $\frac{2x}{\cos \alpha} - \frac{y}{\sin \alpha} = 3$, if it passes through $Q(\beta)$,

$$\text{then } \frac{4 \cos \beta}{\cos \alpha} - \frac{\sin \beta}{\sin \alpha} = 3 \Rightarrow$$

$$\sin \beta = -\sin \alpha \left[\frac{7 + 9 \sin^2 \alpha}{1 + 15 \sin^2 \alpha} \right]$$

3. Ans (C)

$$I = \int_0^{4\pi} \log_e 14 \left| \sin x + \tan^{-1} \left(\frac{3\sqrt{3}}{13} \right) \right| dx \text{ and}$$

$$\int_{a+nT}^{a+mT} f(x) dx = (m-n) \int_0^T f(x) dx, \text{ } f(x) \text{ is a periodic function with period } T$$

4. Ans (D)

Conceptual

5. Ans (C)

$$\text{area} = 4 - \int_0^1 (f(y) - 1) dy + \left| 4 - \int_{-1}^0 (1 - f(y)) dy \right|$$

6. Ans (B)

$$(2x^2 + 1)(y^2 - 3) = 9$$

$$\Rightarrow 2x^2 + 1 = \frac{9}{4^2 - 3} \Rightarrow y = \pm 2 \Rightarrow x = \pm 2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B,C,D)

For positive real numbers, $A.M \geq G.M \Rightarrow k = 200$

8. Ans (A,B,C,D)

Apply integration by parts

9. Ans (B,C,D)

$$f(x) = \frac{\tan x}{x}$$

10. Ans (A,B,C)

$$\text{If } I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx, \text{ then } I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

11. Ans (A,C)

$$PQ^2 = 9PQ$$

12. Ans (B,C,D)

For such points $x = \alpha, f''(\alpha) = 0$

13. Ans (C,D)

$$4(n_{C_1} + 5.n_{C_5} + 9.n_{C_9} + \dots) = n(2^{n-1} + (1+i)^{n-1} + (1-i)^{n-1})$$

14. Ans (B,D)

Conceptual

PART-3 : MATHEMATICS

SECTION-I (iii)

15. Ans (A)

Graph of $y = g(x)$ is

16. Ans (B)

Graph of $y = g(x)$ is

17. Ans (A)

Conceptual

18. Ans (A)

Conceptual